A Quick Look at Multi-architecture, Multi-platform Mobile App Development Frameworks

The article presents an overview of multi-architecture, multi-platform mobile application development frameworks and then takes a look at appEZ, which, as its website proudly declares, is "Ridiculously easy mobile app building for non-programmers."

The pace at which the mobile is changing people's life styles is accelerating. The emerging challenge for enterprise applications is to be available and accessible over diverse mobile platforms. Multi-architecture (flexibility to support native, hybrid and Web mobile applications) and multi-platform (Android, iOS, WP, etc) support is the imperative expectation for any mobile application development tool/framework. Along with cross-platform support, such a framework is also expected to help mobile applications cover platform-specific diverse form factors. Such diversity among mobile devices and the rich feature support in HTML/CSS have resulted in mobile hybrid applications being a preferred implementation approach.

Mobile applications can be classified into three types - native mobile applications, Web applications and hybrid mobile applications.

Native mobile applications

These are also known as thick client applications. These apps are implemented via Android, iOS, Windows Phone or other mobile OSs. The imperative characteristics of any native mobile application are:

- An executable file installs and resides at the mobile device
- Is executed directly by the mobile operating system
- Is able to use the mobile platform or operating system APIs
- Is distributed via a platform-specific app store or via an enterprise distribution mechanism

Mobile Web applications

These are also known as thin client applications. These apps are implemented with Web technologies (HTML, CSS and JavaScript). Some imperative characteristics for mobile Web applications are:

- Applications are executed by the device browser of the mobile operating system
- They can leverage only limited device features for app implementation
- They don't carry any executable file that can be installed or removed from the mobile OS

Mobile hybrid applications

Mobile hybrid apps are neither native nor Web applications. They are implemented with Web technologies and packaged as applications for distribution. These applications can access native device features and APIs. Basically, hybrid applications are native mobile applications that host Web browser controls within their main UI screen. Here are the imperative